FINM3013 Introduction to Financial Derivatives
Quiz 1

Name: _____ Student ID: _____

Date: 21 March 2025
Time allowed: 60 minutes **Full mark: 40**

I. Multiple choice questions (10 points):

*Identify the letter of the choice that **best** completes the statement or answers the question. 2 points for each question.*

1. In the currency futures market, what is the term that describes the number of contracts in a currency outstanding at any one time?
   A. open interest
   B. closed interest
   C. long position
   D. short position

2. Which of the following is correct? The major disadvantage of forward and futures contracts relative to options is that forwards and futures contracts
   A. cannot protect the holder against the risk of adverse movements
   B. are more expensive
   C. are available only for relatively short maturities
   D. eliminate the possibility of gaining a windfall profit from favourable movements

3. An item shows how much money must be in the futures market account balance when the contract is first entered into. It is the
   A. initial margin
   B. margin call
   C. maintenance margin
   D. account balance

4. Suppose you are holding a long position in a euro futures contract that matures in 76 days. The agreed-upon price is $1.15 for 125,000 euro. At the close of trading today, the futures price has risen to $1.155. Under daily settlement, you now
   A. hold a futures contract that has risen in value by $1,250
   B. hold a futures contract that has fallen in value by $625
   C. will receive $625 and a new futures contract priced at $1.155
   D. must pay over $1,250 to the seller of the futures contract

5. A company has a $36 million portfolio with a beta of 1.2. The futures price for a contract on an index is 900. Futures contracts on $250 times the index can be traded. What trade is necessary to increase beta to 1.5?
   A. Long 192 contracts
   B. Short 192 contracts
   C. Long 48 contracts
   D. Short 48 contracts

II. Problems (30 points)

1. (12 points)

A trader invests in Facebook by buying 1000 shares in June for $30 per share. She also buys 1000 put options for $5 each as insurance in case the stock drops sharply. The put options have a strike price of $30 and a maturity date of December. What is the gain or loss if the spot price on December is: a) $20, b) $30, c) $35 and d) $37?

2. (8 points)

A company enters into a long futures contract to buy 5,000 bushels of wheat for 200 cents per bushel. The initial margin is $3,000 and the maintenance margin is $2,000. What price change would lead to a margin call? Under what circumstances could $1,500 be withdrawn from the margin account?

3. (10 points)

Suppose that the spot price of a stock is $17 and the quoted one-year forward price is $18. The risk free interest rate is $r = 8\%$ per annum with annual compounding. Assume that short selling requires a 30% security deposit attracting interest at $d = 4\%$ per annum with annual compounding. Is there an arbitrage opportunity? Find the highest rate d for which there is no arbitrage opportunity.